CODE SPORT

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In this month's column, we take a break from our discussion on natural language processing (NLP) to feature a set of questions in computer science for our student readers.

In the last few columns, we have been discussing different aspects of NLP. This month, we feature a set of interview questions. While the questions generally cover the areas of algorithms and operating systems, I was requested by a couple of students to feature questions relating to machine learning, data science and NLP. So this month's question set features questions from all these areas.

1. Most of you will be familiar with Netflix's contest, in which the company offered a prize for a system that could suggest movies automatically for a user. Basically, given a movie and a particular user, it asked you to predict what would be the user's ratings for that movie. Contestants were given 480,189 users who had rated 17,770 movies. The training data set consisted of tuples of the form (user ID, movie ID, date of grade, grade given). For the test data set which consisted of tuples of the form (user ID, movie ID, date of grade), contestants had to predict the rating for each tuple in the test data set.
   a) Can you come up with an algorithm that would be able to do the prediction?
   b) The above problem belongs to the domain known as 'recommender systems', which applies to many domains including e-commerce, health care, education, etc. Let's assume that you have designed an algorithm for the above problem of movie ratings which works well. Now you are given a data set consisting of educational courses and users. The training data consists of the ratings given by the users for the courses. Just as in the earlier problem, you are asked to predict the rating for a course in the test data set for a given user. Would you be able to use the solution you created for question (a), as it is? If not, what are the changes that you would need to make?
   c) Consider problem (a) again, for which you are given large amounts of training data. But consider a scenario in which you have just started a movie rental company. So you have only 1000 samples of training data. In that case, how can you handle the problem 'cold start'?
2. In question (1) above, we considered the problem of collaborative filtering, in which users who have rated certain items in the past are now given suggestions on what items they might like in the future. Basically, consider the rating values represented in a two dimensional matrix of the form (Users X Movies), where each user is represented by a row in the movie and each column in the matrix stands for a movie. In the problem given above, the matrix is of the form (480,189 X 17,770) elements.
   - Given that each matrix element takes 4 bytes (an integer rating value), what would be the amount of memory needed to hold this matrix in memory?
   - Since each user would have watched only some of the movies among the total number of 17,740, he/she can provide ratings only for a few movies. Let us assume that a non-rated movie is given a special value of 0 whereas the valid ratings are from 1 to 5. In such a case, most of the matrix elements would be zero. So instead of occupying (480,189 X 17,770 X 4) bytes in memory, can you come up with an efficient way of storing the sparse matrix in memory and manipulating it?
3. What is A/B testing? If you are designing a website, how would you use A/B testing?
4. Most of you would be familiar with Map Reduce architecture and Hadoop, which is the most popular implementation of the former. Why is Map Reduce known as 'Shared Nothing' architecture? Can you give a high level overview of Hadoop architecture?
5. Following up on the above question, what are the performance bottlenecks associated with a traditional Hadoop cluster? In other words, is it CPU bound, memory bound or I/O bound? Consider that you have a Map Reduce application running on a four-node Hadoop cluster. You are given the following options: